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6.2 Cumulative Frequency, Histograms and Box Plots

OCR GCSE Mathematics J560 — Unit 6 — Papers 2 and 3 (Higher content marked)

6.2.1 Cumulative Frequency Tables and Graphs

Cumulative frequency is a running total of frequencies up to (and including) the end of each class interval.

The cumulative frequency graph (ogive) is plotted with:

- **x-axis**: the *upper boundary* of each class interval
- **y-axis**: the cumulative frequency

Points are joined with a smooth S-shaped curve. The curve always starts at the lower boundary of the first class (cumulative frequency = 0) and ends at the total frequency.

Worked Example: The heights of 60 plants are recorded.

Height h (cm)	Frequency	Cumulative Frequency
$0 \leq h < 10$	5	5
$10 \leq h < 20$	12	17
$20 \leq h < 30$	20	37
$30 \leq h < 40$	15	52

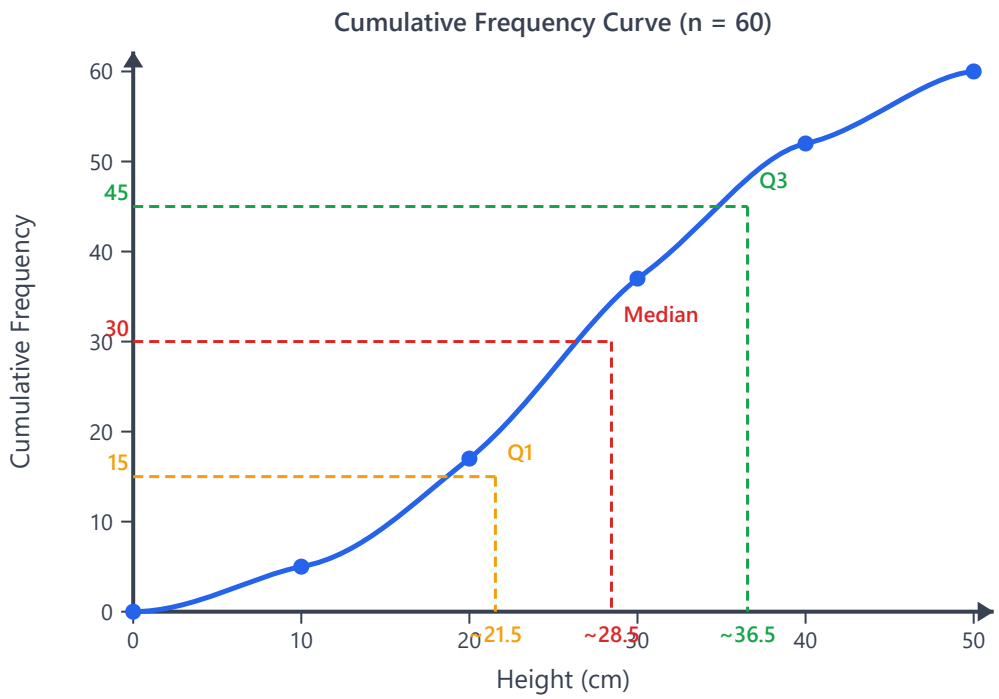
$40 \leq h < 50$	8	60

Plot points at: $(10, 5)$, $(20, 17)$, $(30, 37)$, $(40, 52)$, $(50, 60)$, and also $(0, 0)$ as the starting point.

Reading Median and Quartiles from the Curve

For a total frequency n:

- **Median**: read off at cumulative frequency $= \frac{n}{2}$
- **Lower Quartile (Q1)**: read off at cumulative frequency $= \frac{n}{4}$
- **Upper Quartile (Q3)**: read off at cumulative frequency $= \frac{3n}{4}$
- **Interquartile Range (IQR)** $= Q3 - Q1$



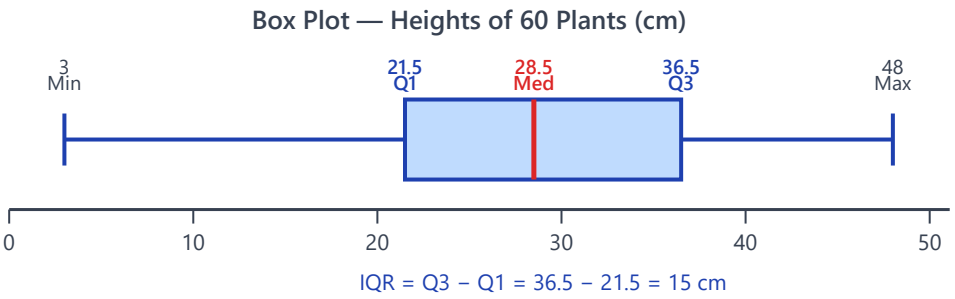
From the graph above: $Q1 \approx 21.5$, Median ≈ 28.5, $Q3 \approx 36.5$, so $\text{IQR} = 36.5 - 21.5 = 15$ cm.

Exam Tip: Always plot cumulative frequency at the **upper class boundary** (not the midpoint). The curve must start at the lower boundary of the first class with CF = 0. Draw the S-curve smoothly — do not join points with ruled straight lines.

6.2.2 Box Plots (Box-and-Whisker Diagrams)

A **box plot** displays the five-number summary of a dataset:

- **Minimum** — smallest value
- **Lower Quartile (Q1)** — 25th percentile
- **Median (Q2)** — 50th percentile (middle value)
- **Upper Quartile (Q3)** — 75th percentile
- **Maximum** — largest value
- **IQR** $= Q3 - Q1$ (the width of the box)



Worked Example: **Constructing a box plot** from the plant height data (Min=3, Q1=21.5, Median=28.5, Q3=36.5, Max=48):

1. Draw a scaled number line.
2. Draw a box from Q1 (21.5) to Q3 (36.5).
3. Draw a vertical line inside the box at the median (28.5).
4. Draw whiskers from Q1 to the minimum (3) and from Q3 to the maximum (48).
5. Add short vertical lines at the minimum and maximum.

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$\text{IQR} = 36.5 - 21.5 = 15$ cm

Exam Tip: The whiskers go to the **minimum and maximum** values in the data (not to Q1 − 1.5×IQR as in some advanced statistics). In GCSE, always use the actual min and max for the whiskers unless told otherwise.

6.2.3 Histograms [Higher]

Higher Tier Only: Histograms are used for **grouped continuous data with unequal class widths**. Unlike bar charts, the **area** of each bar (not the height) represents frequency.

$$\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}$$

Rearranged: $\text{Frequency} = \text{Frequency Density} \times \text{Class Width}$

Worked Example: The times taken (in minutes) by 80 students to complete a task are recorded.

Time t (min)	Frequency	Class Width	Frequency Density
$0 \leq t < 10$	8	10	$8 \div 10 = 0.8$
$10 \leq t < 20$	20	10	$20 \div 10 = 2.0$
$20 \leq t < 30$	24	10	$24 \div 10 = 2.4$
$30 \leq t < 50$	20	20	$20 \div 20 = 1.0$
$50 \leq t < 80$	8	30	$8 \div 30 \approx 0.27$

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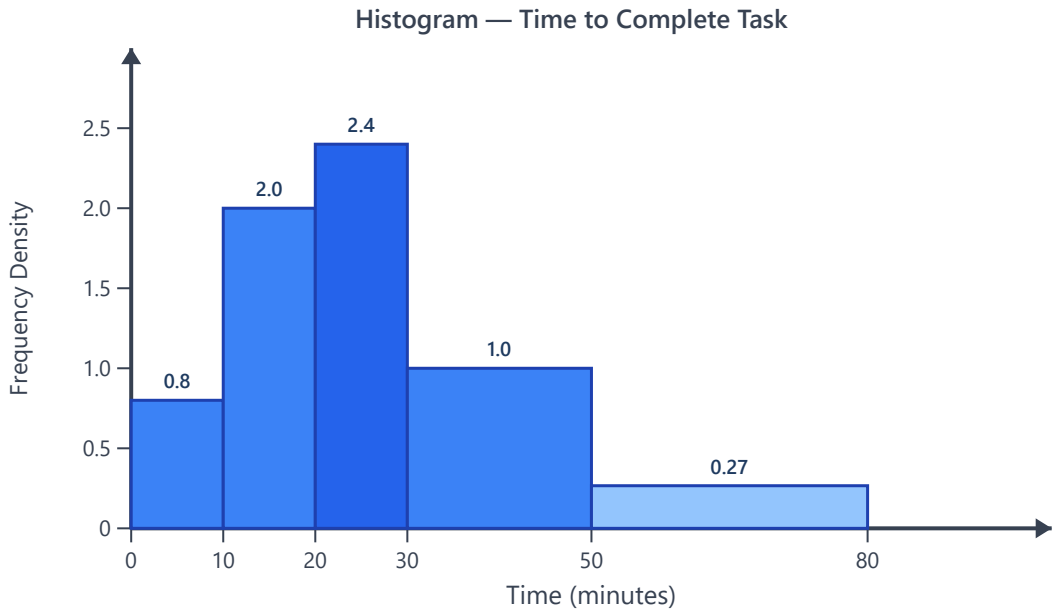
Note: The last two classes have *wider* class widths. Without using frequency density, those bars would appear deceptively tall.



Reading Frequency from a Histogram

Worked Example: Using the histogram above: how many students took between 30 and 50 minutes?

Bar from 30 to 50: class width = 20, frequency density = 1.0

$\text{Frequency} = 1.0 \times 20 = 20$ students

Estimating the Number in Part of a Class

Worked Example: How many students took between 20 and 25 minutes?

Bar from 20 to 30: FD = 2.4, class width = 10

We want only half the bar (20 to 25): class width = 5

$\text{Frequency} = 2.4 \times 5 = 12$ students

This assumes the data is evenly spread within the class interval.

Exam Tip: The y-axis of a histogram is always **frequency density**, never frequency. Never label it as frequency. Always calculate frequency density = frequency ÷ class width before drawing. The area of each bar = frequency.

6.2.4 Comparing Distributions

To compare two distributions you must comment on **both**:

- A measure of **average** (usually median when using box plots or cumulative frequency)
- A measure of **spread** (usually IQR when using box plots; range if appropriate)

Always refer back to the **context** of the question.

Worked Example: Two classes took the same test. Their results (out of 100) are summarised below:

Statistic	Class A	Class B
Minimum	30	20
Q1	52	45
Median	65	62
Q3	74	80
Maximum	90	98
IQR	$74 - 52 = 22$	$80 - 45 = 35$

Comparison:

Average: Class A has a higher median (65 vs 62), so Class A performed slightly better on

average.

Spread: Class A has a smaller IQR (22 vs 35), so Class A's scores were more consistent. Class B had more variation in performance.

Interpreting Skewness from a Box Plot

Shape	Box Plot Feature	Meaning
Symmetric	Median in centre of box; whiskers roughly equal	Data evenly distributed around median
Positive skew	Median closer to Q1; longer upper whisker	Most data at lower end; a few high values pull the mean up
Negative skew	Median closer to Q3; longer lower whisker	Most data at upper end; a few low values pull the mean down

Exam Tip: In exam comparison questions, always make **two separate statements**: one about average and one about spread. Each statement must include the actual values (e.g. "The median for Class A is 65, which is higher than Class B's median of 62") and a conclusion in context. Vague answers such as "Class A did better" will not gain full marks.

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